

SAPIENZA
UNIVERSITÀ DI ROMA

Combinatorial Optimization

Exercises 6 — Specialized algorithms

Worked exercises on the chapters of Volume 6 (specialized algorithms), each followed by its complete solution. The instances are those of the «Selected problems» pages, cited by their code.

Fabio Furini

fabiofurini.github.io

Department of Computer, Control and Management Engineering — Sapienza University of Rome

Online version and downloadable material:
fabiofurini.github.io/combinatorial-optimization/
github.com/fabiofurini/combinatorial-optimization

Academic year 2026–2027

Contents

34 Knapsack: branch-and-bound on the critical item **2**

Chapter 34

Knapsack: branch-and-bound on the critical item

Exercise 34.1 — Branch-and-Bound Algorithm for the Knapsack Problem ($P1$)

We consider a knapsack capacity $c = 11$ and a set $\mathscr{I} = \{1, 2, 3, 4, 5\}$ of 5 items ($n = 5$) with the following profits and weights:

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
p_j	8	6	14	6	2
w_j	3	3	9	5	2

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the LIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;
   (d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :

   $$\bar{x}_1 = 1, \quad \bar{x}_2 = 1, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1, \quad \bar{x}_5 = 0$$

   and set the Lower Bound LB accordingly.
3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{8}{3} \geq \frac{6}{3} \geq \frac{14}{9} \geq \frac{6}{5} \geq \frac{2}{2}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{lrlrlrlrlrcl} \max & 8 & x_1 & +6 & x_2 & +14 & x_3 & +6 & x_4 & +2 & x_5 & \\ & 3 & x_1 & +3 & x_2 & +9 & x_3 & +5 & x_4 & +2 & x_5 & \leq & 11 \\ & & x_1, & & x_2, & & x_3, & & x_4, & & x_5 & \in & \{0,1\} \end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{lrlrlrlrlrcl} \max & 8 & x_1 & +6 & x_2 & +14 & x_3 & +6 & x_4 & +2 & x_5 & & \\ & 3 & x_1 & +3 & x_2 & +9 & x_3 & +5 & x_4 & +2 & x_5 & \leq & 11 \\ & & x_1 & & & & & & & & & \leq & 1 \\ & & & & x_2 & & & & & & & \leq & 1 \\ & & & & & & x_3 & & & & & \leq & 1 \\ & & & & & & & & x_4 & & & \leq & 1 \\ & & & & & & & & & & x_5 & \leq & 1 \\ & & x_1, & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0 \end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

  For the LP^1_{KP} problem, the split item is $s = 3$ and the residual capacity is $\bar{c} = 5$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{5}{9}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0$$

  the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 8 \cdot 1 + 6 \cdot 1 + 14 \cdot \frac{5}{9} = \frac{196}{9}$$

  The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 20 ($LB(\bar{\boldsymbol{x}}) = 20$).

  Due to the current Lower Bound LB and Upper Bound UB, we have:

$$20 \quad \leq \quad z(\, ILP\,) \quad \leq \quad \underbrace{\left\lfloor \frac{196}{9} \right\rfloor}_{=21}$$

- **Iteration 1**

  The LP^1_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_3. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

  (a) Solution of the LP^2_{KP} problem

  We have

  $$\mathscr{I}^2 = \{1, 2, 4, 5\}, \quad \mathscr{I}^2_{\text{out}} = \{3\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

  The capacity is $c(2) = 11$ and the objective-function constant is $p(2) = 0$. The split item is $s = 5$ and the residual capacity is $\bar{c} = 0$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

  $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_4 = 1, \quad \tilde{x}_5 = 0, \quad \text{and} \quad z(\, LP^2_{\text{KP}}\,) = 20$$

  The LP^2_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^2_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 20 \leq 20$$

  accordingly, the branching node is pruned by upper bound.

  (b) Solution of the LP^3_{KP} problem

  We have

  $$\mathscr{I}^3 = \{1, 2, 4, 5\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{3\}$$

  The capacity is $c(3) = 2$ and the objective-function constant is $p(3) = 14$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 2$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

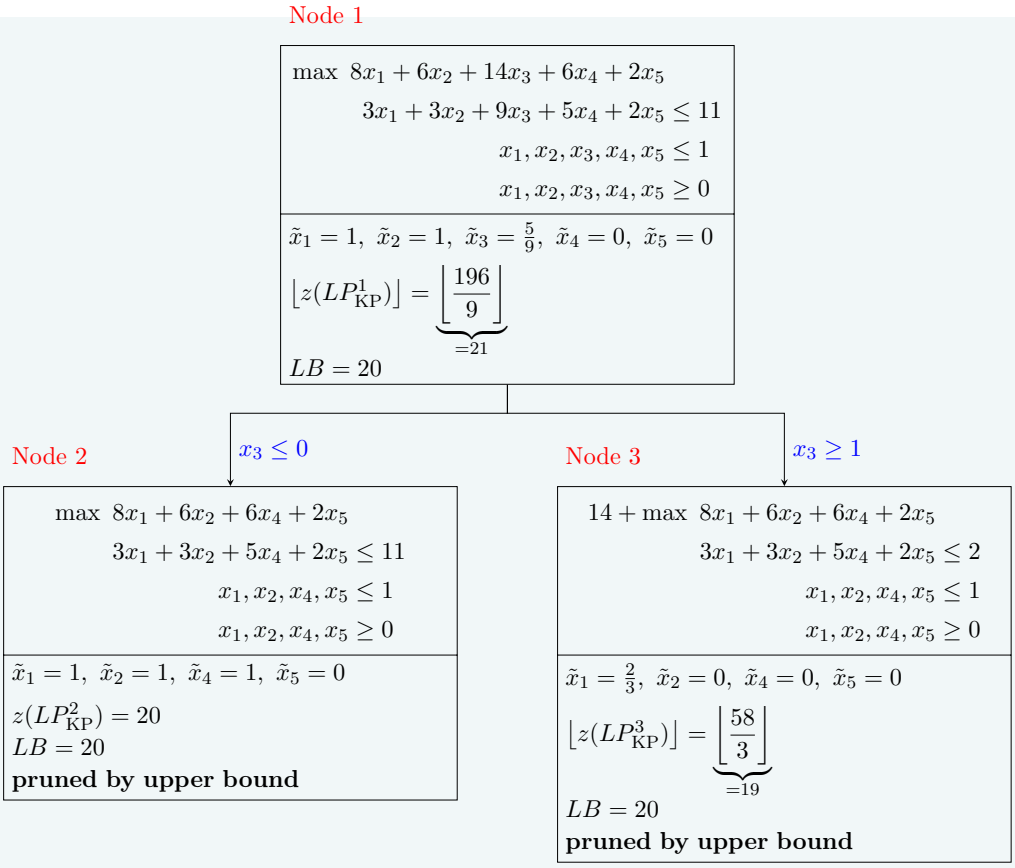
  $$\tilde{x}_1 = \frac{2}{3}, \quad \tilde{x}_2 = 0, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \text{and} \quad z(\, LP^3_{\text{KP}}\,) = 14 + 8 \cdot \frac{2}{3} = \frac{58}{3}$$

  The LP^3_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^3_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{58}{3} \right\rfloor}_{=19} \leq 20$$

  accordingly, the branching node is pruned by upper bound.

  At the end of the first iteration, Node 2 and Node 3 are both pruned by upper bound. The corresponding Branch-and-Bound Tree is:

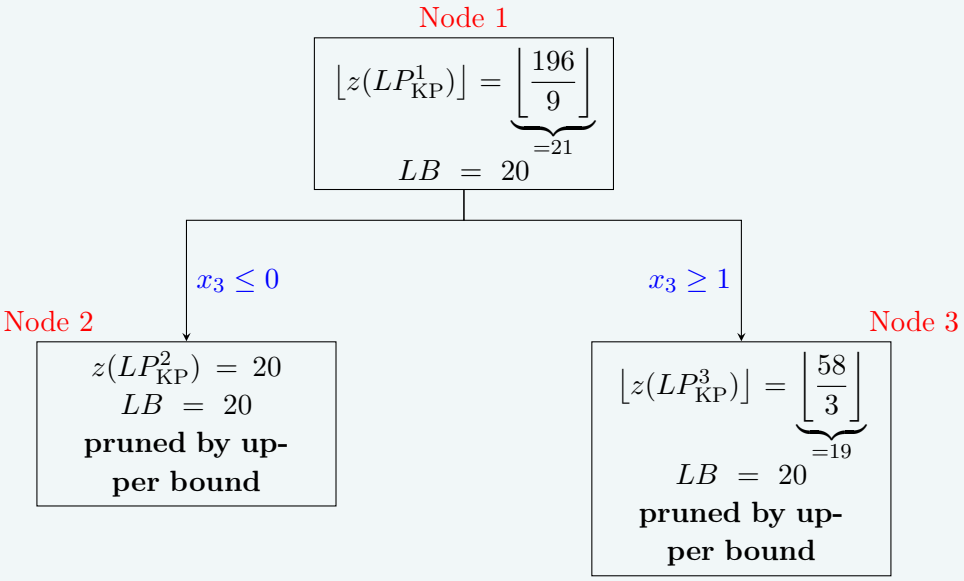


- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 20$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 1, \tilde{x}_3 = 0, \tilde{x}_4 = 1, \tilde{x}_5 = 0$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.



Exercise 34.2 — Branch-and-Bound Algorithm for the Knapsack Problem ($P2$)

We consider a knapsack capacity $c = 8$ and a set $\mathscr{J} = \{1, 2, 3, 4\}$ of 4 items ($n = 4$) with the following profits and weights:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
p_j	40	42	25	12
w_j	4	7	5	3

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the LIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;
   (d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :
   $$\bar{x}_1 = 1, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1$$
   and set the Lower Bound LB accordingly.
3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{40}{4} \geq \frac{42}{7} \geq \frac{25}{5} \geq \frac{12}{3}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 40 & x_1 & +\,42 & x_2 & +\,25 & x_3 & +\,12 & x_4 & & \\ & 4 & x_1 & +\,7 & x_2 & +\,5 & x_3 & +\,3 & x_4 & \leq & 8 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \in & \{0,1\} \end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 40 & x_1 & +\,42 & x_2 & +\,25 & x_3 & +\,12 & x_4 & & \\ & 4 & x_1 & +\,7 & x_2 & +\,5 & x_3 & +\,3 & x_4 & \leq & 8 \\ & & x_1 & & & & & & & \leq & 1 \\ & & & & x_2 & & & & & \leq & 1 \\ & & & & & & x_3 & & & \leq & 1 \\ & & & & & & & & x_4 & \leq & 1 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

  For the LP^1_{KP} problem, the split item is $s = 2$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{4}{7}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0$$

  the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 40 \cdot 1 + 42 \cdot \frac{4}{7} = 64$$

  The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 52 ($LB(\bar{\boldsymbol{x}}) = 52$).

  Due to the current Lower Bound LB and Upper Bound UB, we have:

$$52 \leq z(\, ILP\,) \leq 64$$

- **Iteration 1**

  The LP^1_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_2. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_2 \leq 0$) and selecting ($x_2 \geq 1$) item 2.

  (a) Solution of the LP^2_{KP} problem

  We have

  $$\mathscr{I}^2 = \{1, 3, 4\}, \quad \mathscr{I}^2_{\text{out}} = \{2\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

  The capacity is $c(2) = 8$ and the objective-function constant is $p(2) = 0$. The split item is $s = 3$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

  $$\tilde{x}_1 = 1, \quad \tilde{x}_3 = \frac{4}{5}, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^2_{\text{KP}}\,) = 40 \cdot 1 + 25 \cdot \frac{4}{5} = 60$$

  The LP^2_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^2_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad 60 > 52$$

  The optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^2_{KP} problem is **inserted in the problem list** L.

  (b) Solution of the LP^3_{KP} problem

  We have

  $$\mathscr{I}^3 = \{1, 3, 4\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{2\}$$

  The capacity is $c(3) = 1$ and the objective-function constant is $p(3) = 42$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

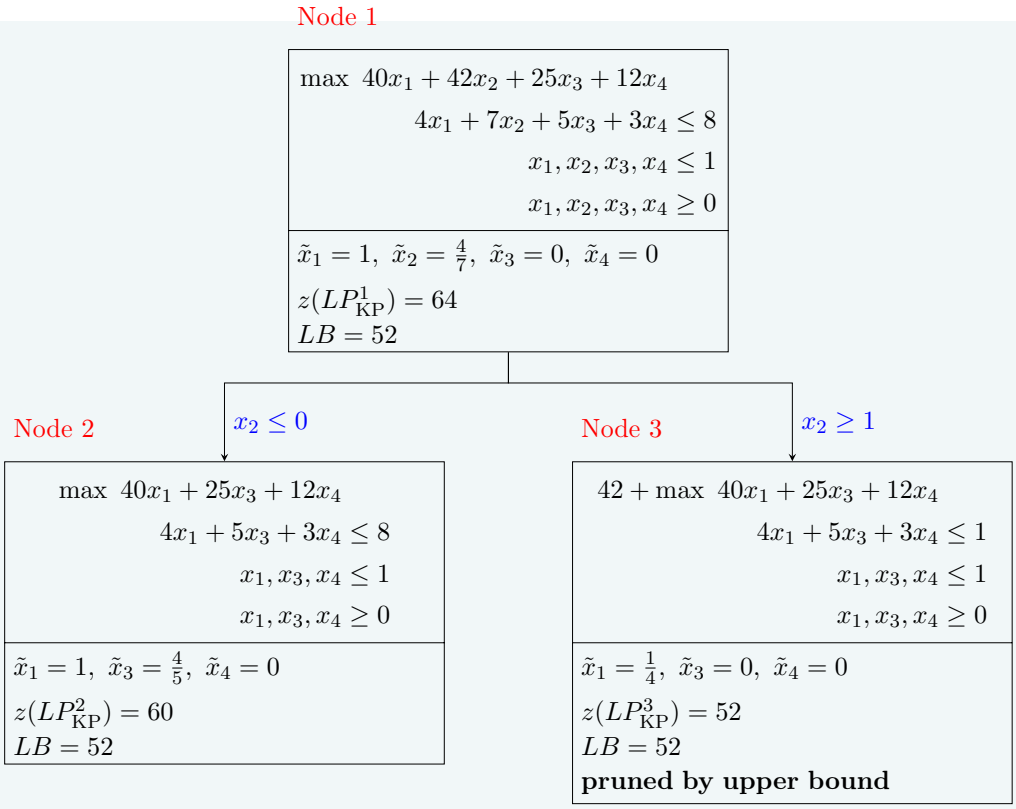
  $$\tilde{x}_1 = \frac{1}{4}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^3_{\text{KP}}\,) = 42 + 40 \cdot \frac{1}{4} = 52$$

  The LP^3_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^3_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 52 \leq 52$$

  accordingly, the branching node is pruned by upper bound.

  At the end of the first iteration, Node 2 is inserted in the Problem List L, while Node 3 is pruned by upper bound. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$52 \quad \leq \quad z(\ ILP\ ) \quad \leq \quad 60$$

- **Iteration 2**

The LP^2_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_3. The LP^4_{KP} and LP^5_{KP} problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

(a) Solution of the LP^4_{KP} problem

We have

$$\mathscr{I}^4 = \{1, 4\}, \quad \mathscr{I}^4_{\text{out}} = \{2, 3\}, \quad \mathscr{I}^4_{\text{in}} = \emptyset$$

The capacity is $c(4) = 8$ and the objective-function constant is $p(4) = 0$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\ LP^4_{\text{KP}}\ )$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_4 = 1 \quad \text{and} \quad z(\ LP^4_{\text{KP}}\ ) = 40 \cdot 1 + 12 \cdot 1 = 52$$

The LP^4_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^4_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 52 \leq 52$$

accordingly, the branching node is pruned by upper bound.

(b) Solution of the LP^5_{KP} problem

We have

$$\mathscr{I}^5 = \{1, 4\}, \quad \mathscr{I}^5_{\text{out}} = \{2\}, \quad \mathscr{I}^5_{\text{in}} = \{3\}$$

The capacity is $c(5) = 3$ and the objective-function constant is $p(5) = 25$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 3$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^5_{\text{KP}}\,)$ is:

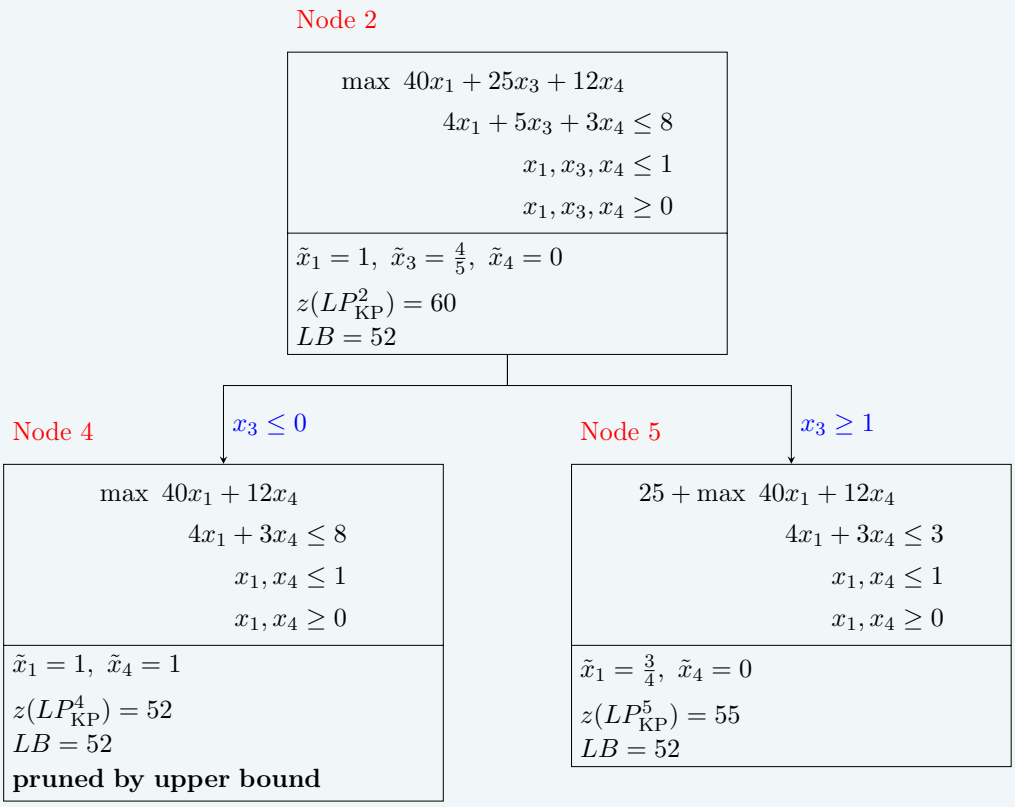
$$\tilde{x}_1 = \frac{3}{4}, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 1, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^5_{\text{KP}}\,) = 25 + 40 \cdot \frac{3}{4} = 55$$

The LP^5_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^5_{\text{KP}}) \right\rfloor > LB \qquad \text{since} \qquad 55 > 52$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^5_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^5_{KP} problem is **inserted in the problem list** L.

At the end of the second iteration, Node 5 is inserted in the Problem List L, while Node 4 is pruned by upper bound. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$52 \quad \leq \quad z(\, ILP\,) \quad \leq \quad 55$$

- **Iteration 3**

  The LP^5_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_1. The LP^6_{KP} and LP^7_{KP} problems are created by unselecting ($x_1 \leq 0$) and selecting ($x_1 \geq 1$) item 1.

  (a) Solution of the LP^6_{KP} problem

  We have

  $$\mathscr{I}^6 = \{4\}, \quad \mathscr{I}^6_{\text{out}} = \{1, 2\}, \quad \mathscr{I}^6_{\text{in}} = \{3\}$$

  The capacity is $c(6) = 3$ and the objective-function constant is $p(6) = 25$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^6_{\text{KP}}\,)$ is:

  $$\tilde{x}_4 = 1 \quad \text{and} \quad z(\, LP^6_{\text{KP}}\,) = 25 + 12 \cdot 1 = 37$$

  The LP^6_{KP} problem is **feasible** and we have:

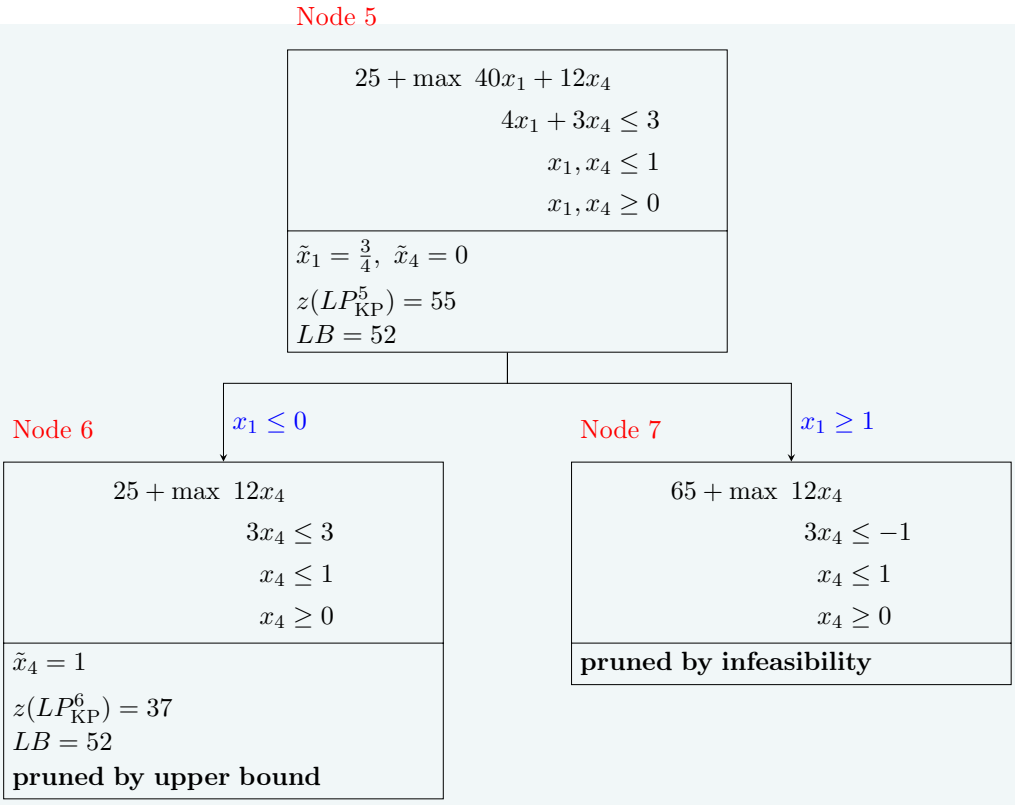
  $$\left\lfloor z(LP^6_{\text{KP}}) \right\rfloor \leq LB \qquad \text{since} \qquad 37 \leq 52$$

  accordingly, the branching node is pruned by upper bound.

  (b) Solution of the LP^7_{KP} problem

  The LP^7_{KP} problem is **infeasible**, accordingly the branching node is pruned by infeasibility.

  At the end of the third iteration, Node 6 is pruned by upper bound and Node 7 is pruned by infeasibility. The corresponding Branch-and-Bound Tree is:

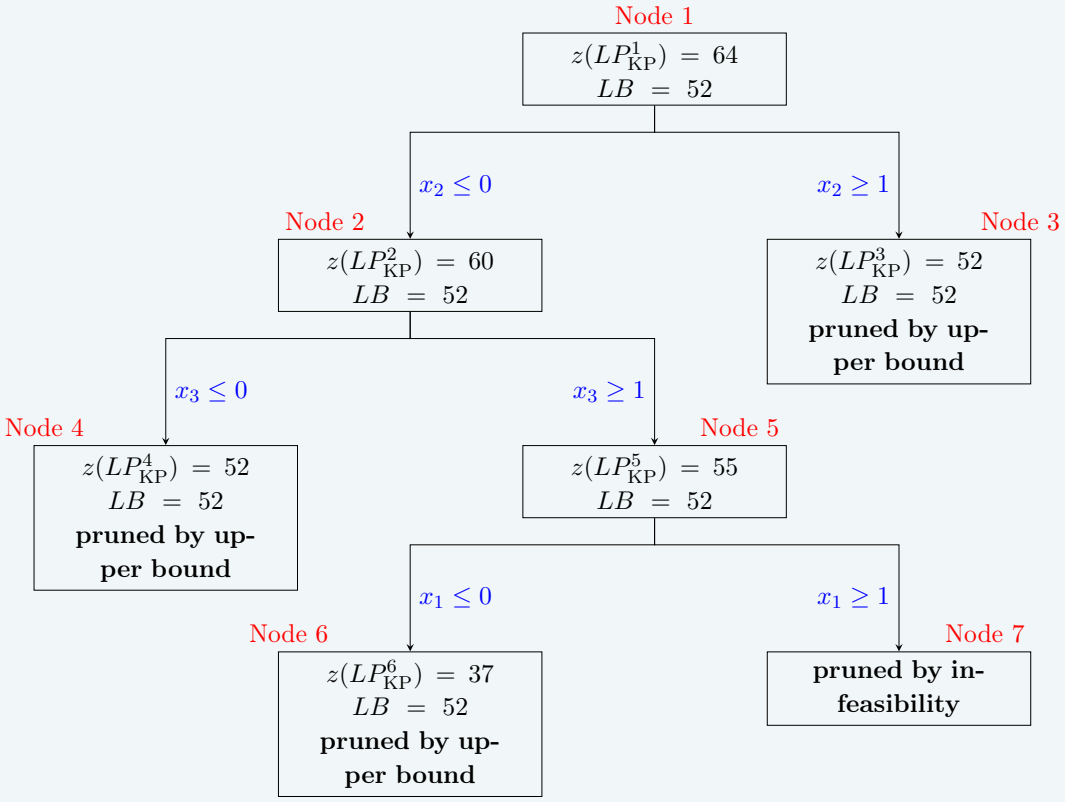


- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 52$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.



Exercise 34.3 — Branch-and-Bound Algorithm for the Knapsack Problem ($P3$)

We consider a knapsack capacity $c = 15$ and a set $\mathscr{I} = \{1, 2, 3, 4\}$ of 4 items ($n = 4$) with the following profits and weights:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
p_j	24	17	12	6
w_j	10	8	6	5

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the LIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;

(d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :

$$\bar{x}_1 = 1, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1$$

and set the Lower Bound LB accordingly.

3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{24}{10} \geq \frac{17}{8} \geq \frac{12}{6} \geq \frac{6}{5}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 24 & x_1 & +\,17 & x_2 & +\,12 & x_3 & +\,6 & x_4 & & \\ & 10 & x_1 & +\,8 & x_2 & +\,6 & x_3 & +\,5 & x_4 & \leq & 15 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \in & \{0,1\} \end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 24 & x_1 & +\,17 & x_2 & +\,12 & x_3 & +\,6 & x_4 & & \\ & 10 & x_1 & +\,8 & x_2 & +\,6 & x_3 & +\,5 & x_4 & \leq & 15 \\ & & x_1 & & & & & & & \leq & 1 \\ & & & & x_2 & & & & & \leq & 1 \\ & & & & & & x_3 & & & \leq & 1 \\ & & & & & & & & x_4 & \leq & 1 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

  For the LP^1_{KP} problem, the split item is $s = 2$ and the residual capacity is $\bar{c} = 5$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{5}{8}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0$$

  the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 24 \cdot 1 + 17 \cdot \frac{5}{8} = \frac{277}{8}$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 30 ($LB(\bar{\boldsymbol{x}}) = 30$).

Due to the current Lower Bound LB and Upper Bound UB, we have:

$$30 \quad \leq \quad z(\, ILP\,) \quad \leq \quad \underbrace{\left\lfloor \frac{277}{8} \right\rfloor}_{=34}$$

- **Iteration 1**

The LP^1_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_2. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_2 \leq 0$) and selecting ($x_2 \geq 1$) item 2.

(a) Solution of the LP^2_{KP} problem

We have

$$\mathscr{I}^2 = \{1, 3, 4\}, \quad \mathscr{I}^2_{\text{out}} = \{2\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

The capacity is $c(2) = 15$ and the objective-function constant is $p(2) = 0$. The split item is $s = 3$ and the residual capacity is $\bar{c} = 5$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_3 = \frac{5}{6}, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^2_{\text{KP}}\,) = 24 \cdot 1 + 12 \cdot \frac{5}{6} = 34$$

The LP^2_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^2_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad 34 > 30$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^2_{KP} problem is **inserted in the problem list** L.

(b) Solution of the LP^3_{KP} problem

We have

$$\mathscr{I}^3 = \{1, 3, 4\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{2\}$$

The capacity is $c(3) = 7$ and the objective-function constant is $p(3) = 17$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 7$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

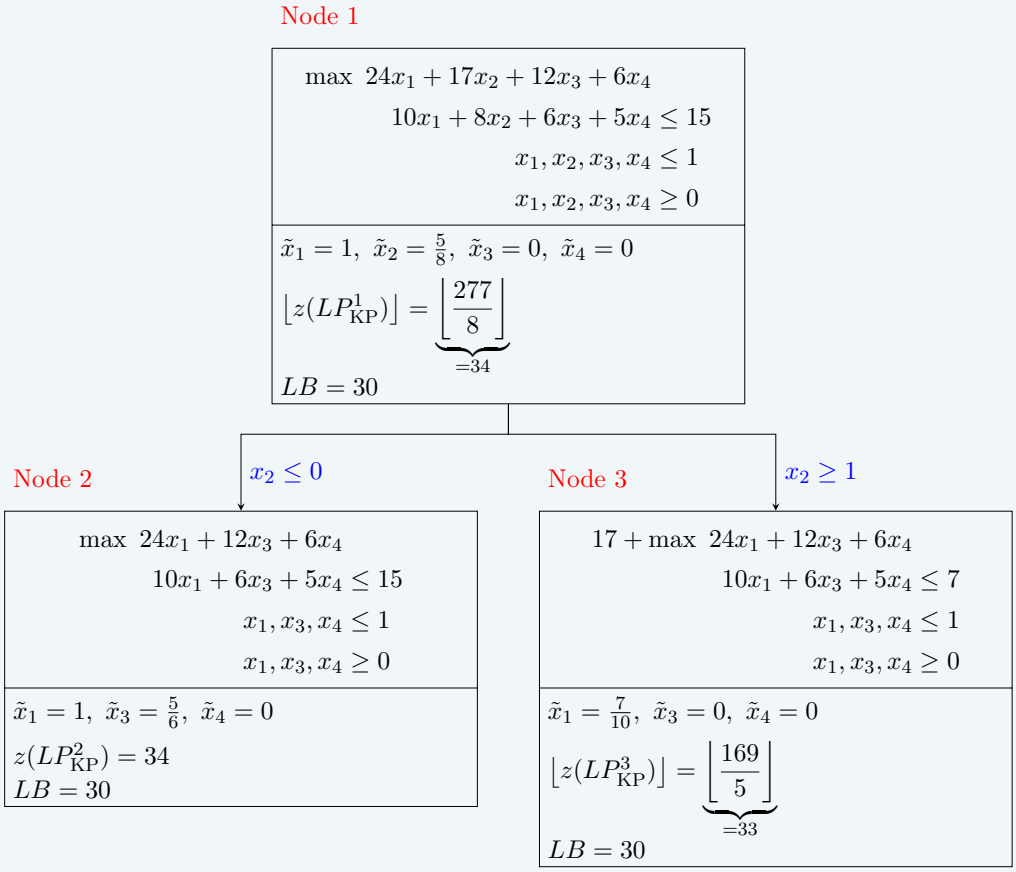
$$\tilde{x}_1 = \frac{7}{10}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^3_{\text{KP}}\,) = 17 + 24 \cdot \frac{7}{10} = \frac{169}{5}$$

The LP^3_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^3_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{169}{5} \right\rfloor}_{=33} > 30$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^3_{KP} problem is **inserted in the problem list** L.

At the end of the first iteration, Node 2 and Node 3 are both inserted in the Problem List L. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$30 \;\leq\; z(\; ILP\;) \;\leq\; \underbrace{\max\left\{34,\; \underbrace{\left\lfloor \frac{169}{5} \right\rfloor}_{=33}\right\}}_{=34}$$

- **Iteration 2**

The LP^3_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_1. The LP^4_{KP} and LP^5_{KP} problems are created by unselecting ($x_1 \leq 0$) and selecting ($x_1 \geq 1$) item 1.

(a) Solution of the LP^4_{KP} problem

We have

$$\mathscr{I}^4 = \{3, 4\}, \quad \mathscr{I}^4_{\text{out}} = \{1\}, \quad \mathscr{I}^4_{\text{in}} = \{2\}$$

The capacity is $c(4) = 7$ and the objective-function constant is $p(4) = 17$. The split item is $s = 4$ and the residual capacity is $\bar{c} = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^4_{\text{KP}}\,)$ is:

$$\tilde{x}_3 = 1, \quad \tilde{x}_4 = \frac{1}{5} \quad \text{and} \quad z(\, LP^4_{\text{KP}}\,) = 17 + 12 \cdot 1 + 6 \cdot \frac{1}{5} = \frac{151}{5}$$

The LP^4_{KP} problem is **feasible** and we have:

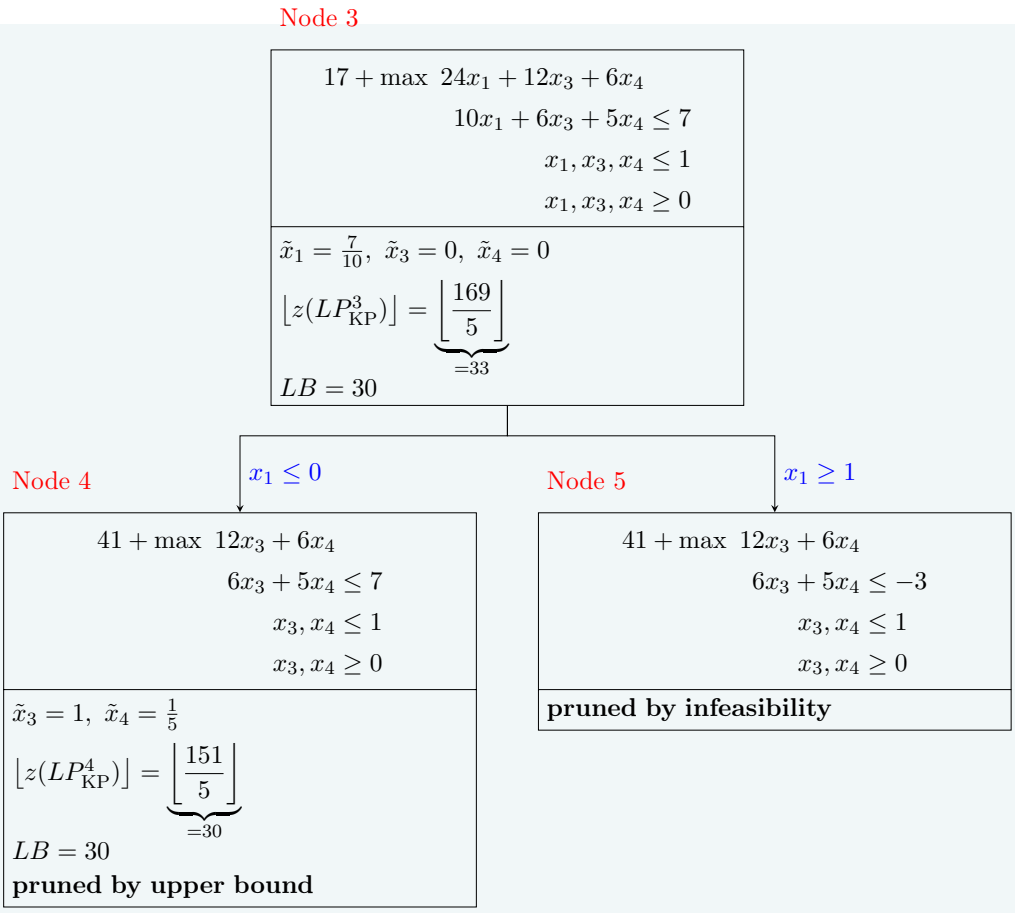
$$\lfloor z(LP^4_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{151}{5} \right\rfloor}_{=30} \leq 30$$

accordingly, the branching node is pruned by upper bound.

(b) Solution of the LP^5_{KP} problem

The LP^5_{KP} problem is **infeasible**, accordingly the branching node is pruned by infeasibility.

At the end of the second iteration, Node 4 is pruned by upper bound and Node 5 is pruned by infeasibility. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$30 \quad \leq \quad z(\ ILP\ ) \quad \leq \quad 34$$

- **Iteration 3**

The LP^2_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_3. The LP^6_{KP} and LP^7_{KP} problems are created by unselecting $(x_3 \leq 0)$ and selecting $(x_3 \geq 1)$ item 3.

(a) Solution of the LP^6_{KP} problem

We have

$$\mathcal{I}^6 = \{1, 4\}, \quad \mathcal{I}^6_{\text{out}} = \{2, 3\}, \quad \mathcal{I}^6_{\text{in}} = \emptyset$$

The capacity is $c(6) = 15$ and the objective-function constant is $p(6) = 0$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\ LP^6_{\text{KP}}\ )$ is:

$$\tilde{x}_1 = 1, \tilde{x}_4 = 1 \quad \text{and} \quad z(\ LP^6_{\text{KP}}\ ) = 24 \cdot 1 + 6 \cdot 1 = 30$$

The LP^6_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^6_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 30 \leq 30$$

accordingly, the branching node is pruned by upper bound.

(b) Solution of the LP^7_{KP} problem

We have

$$\mathscr{I}^7 = \{1, 4\}, \quad \mathscr{I}^7_{\text{out}} = \{2\}, \quad \mathscr{I}^7_{\text{in}} = \{3\}$$

The capacity is $c(7) = 9$ and the objective-function constant is $p(7) = 12$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 9$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^7_{\text{KP}}\,)$ is:

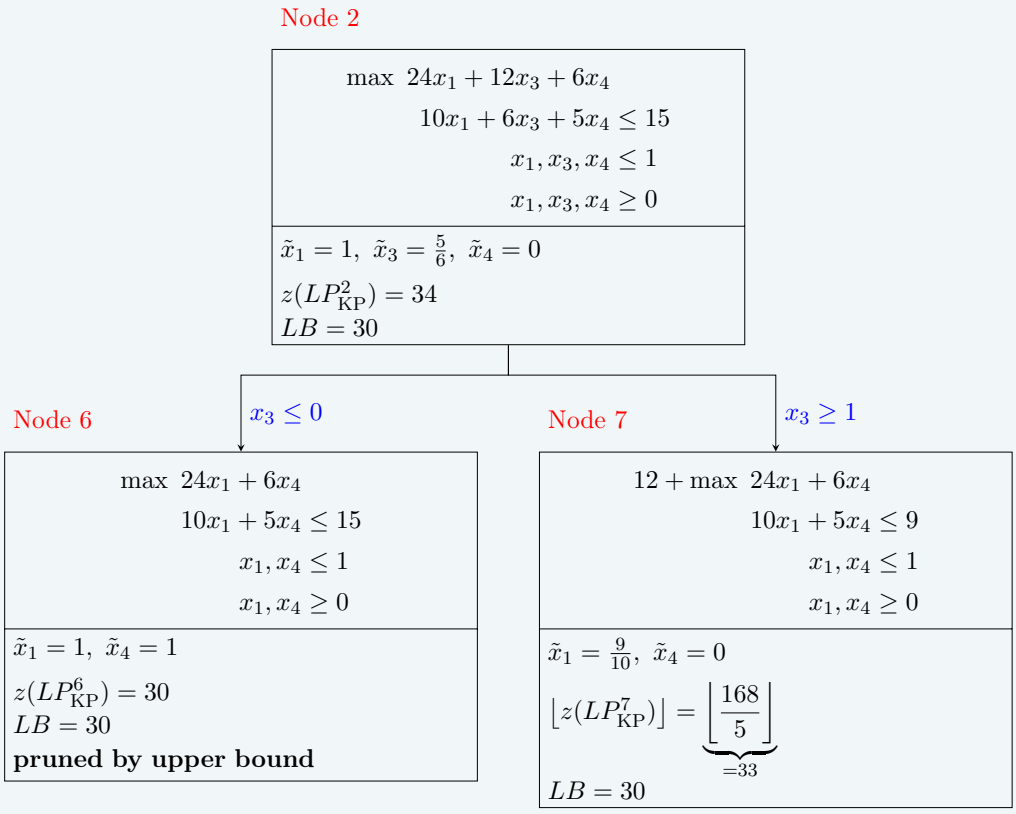
$$\tilde{x}_1 = \frac{9}{10}, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^7_{\text{KP}}\,) = 12 + 24 \cdot \frac{9}{10} = \frac{168}{5}$$

The LP^7_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^7_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{168}{5} \right\rfloor}_{=33} > 30$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^7_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^7_{KP} problem is **inserted in the problem list** L.

At the end of the third iteration, Node 6 is pruned by upper bound and Node 7 is inserted in the Problem List L. The corresponding Branch-and-Bound Tree is:



- **Iteration 4**

  The LP^7_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_1. The LP^8_{KP} and LP^9_{KP} problems are created by unselecting ($x_1 \leq 0$) and selecting ($x_1 \geq 1$) item 1.

  (a) Solution of the LP^8_{KP} problem

  We have

  $$\mathscr{I}^8 = \{4\}, \quad \mathscr{I}^8_{\text{out}} = \{1, 2\}, \quad \mathscr{I}^8_{\text{in}} = \{3\}$$

  The capacity is $c(8) = 9$ and the objective-function constant is $p(8) = 12$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^8_{\text{KP}}\,)$ is:

  $$\tilde{x}_4 = 1 \quad \text{and} \quad z(\, LP^8_{\text{KP}}\,) = 12 + 6 \cdot 1 = 18$$

  The LP^8_{KP} problem is **feasible** and we have:

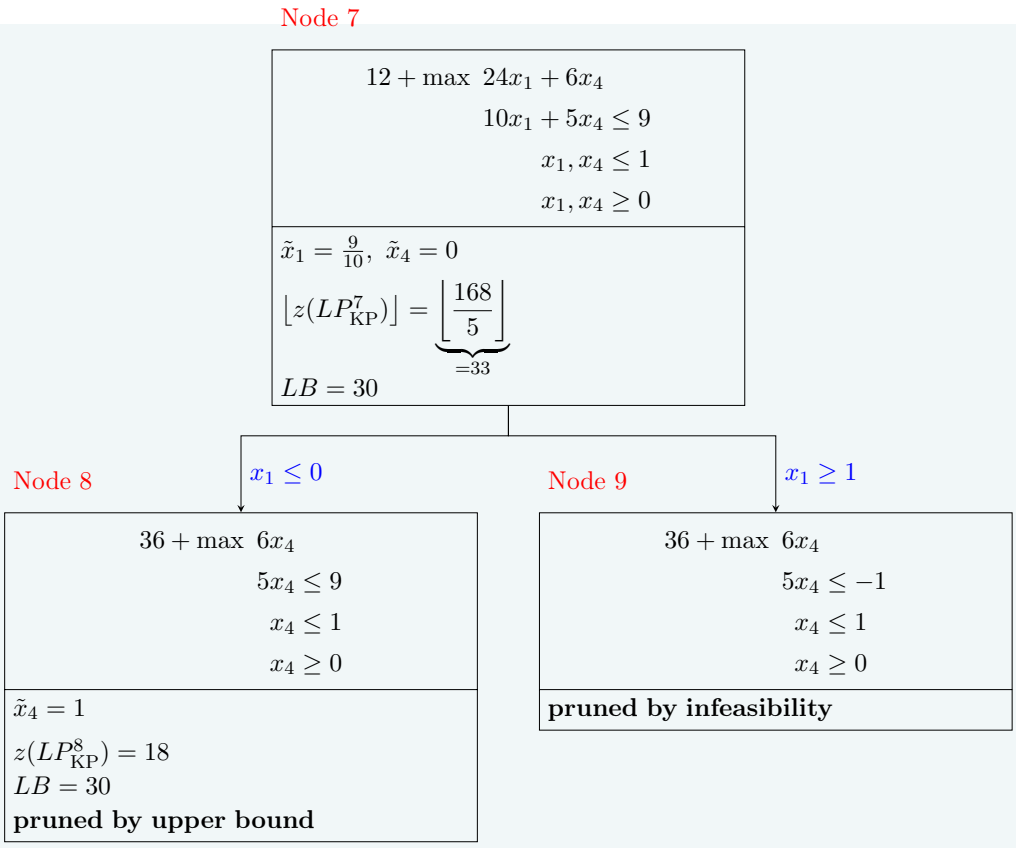
  $$\left\lfloor z(LP^8_{\text{KP}}) \right\rfloor \leq LB \qquad \text{since} \qquad 18 \leq 30$$

  accordingly, the branching node is pruned by upper bound.

  (b) Solution of the LP^9_{KP} problem

  The LP^9_{KP} problem is **infeasible**, accordingly the branching node is pruned by infeasibility.

  At the end of the fourth iteration, Node 8 is pruned by upper bound and Node 9 is pruned by infeasibility. The corresponding Branch-and-Bound Tree is:

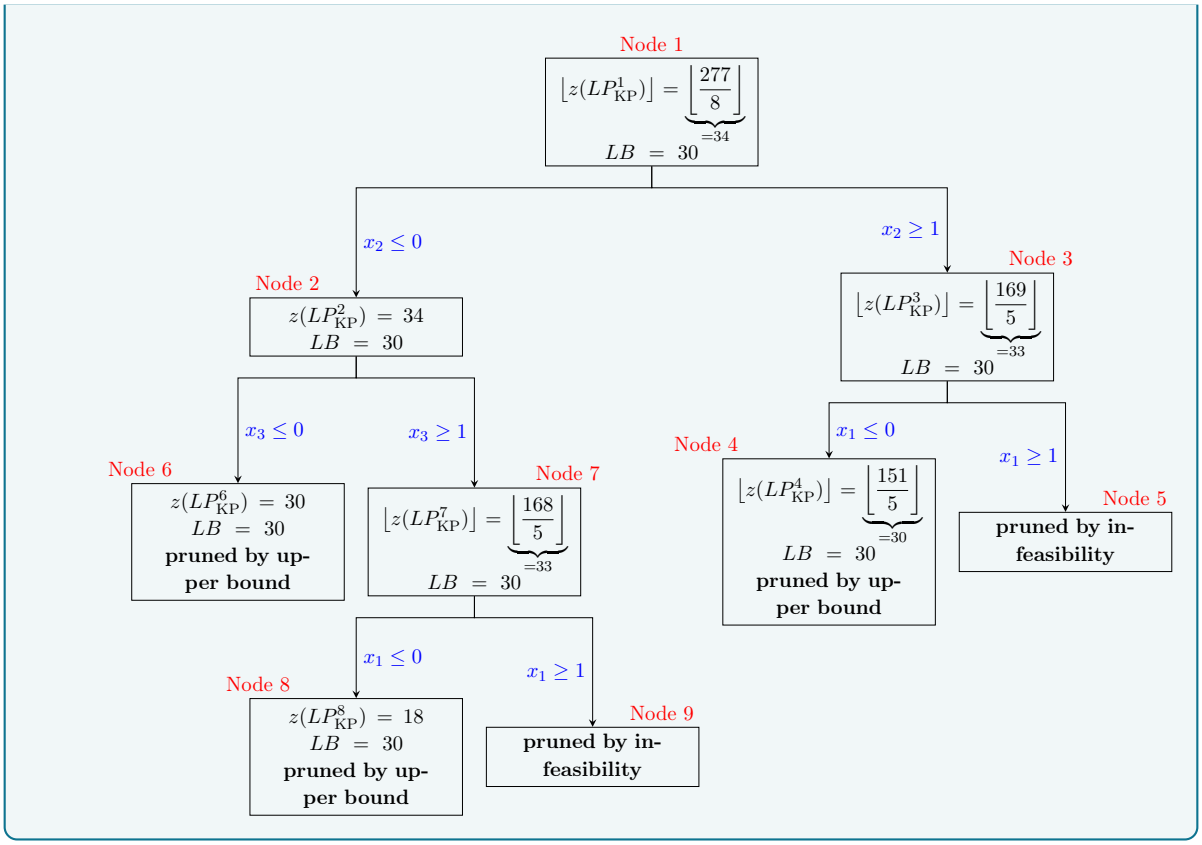


- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 30$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.



Exercise 34.4 — Branch-and-bound Algorithm for the Knapsack Problem ($P4$)

We consider a knapsack capacity $c = 9$ and a set $\mathscr{I} = \{1, 2, 3, 4, 5, 6\}$ of 6 items ($n = 6$) with the following profits and weights:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$
p_i	6	5	8	9	6	7
w_i	2	3	6	7	5	9

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the FIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;

(d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :

$$\bar{x}_1 = 1, \quad \bar{x}_2 = 1, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 0, \quad \bar{x}_5 = 0, \quad \bar{x}_6 = 0$$

and set the Lower Bound LB accordingly.

3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{6}{2} \geq \frac{5}{3} \geq \frac{8}{6} \geq \frac{9}{7} \geq \frac{6}{5} \geq \frac{7}{9}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrrrrrr}
\max & 6 & x_1 & +5 & x_2 & +8 & x_3 & +9 & x_4 & +6 & x_5 & +7 & x_6 & & \\
& 2 & x_1 & +3 & x_2 & +6 & x_3 & +7 & x_4 & +5 & x_5 & +9 & x_6 & \leq & 9 \\
& & x_1, & & x_2, & & x_3, & & x_4, & & x_5, & & x_6 & \in & \{0,1\}
\end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrrrrrr}
\max & 6 & x_1 & +5 & x_2 & +8 & x_3 & +9 & x_4 & +6 & x_5 & +7 & x_6 & & \\
& 2 & x_1 & +3 & x_2 & +6 & x_3 & +7 & x_4 & +5 & x_5 & +9 & x_6 & \leq & 9 \\
& & x_1 & & & & & & & & & & & \leq & 1 \\
& & & & x_2 & & & & & & & & & \leq & 1 \\
& & & & & & x_3 & & & & & & & \leq & 1 \\
& & & & & & & & x_4 & & & & & \leq & 1 \\
& & & & & & & & & & x_5 & & & \leq & 1 \\
& & & & & & & & & & & & x_6 & \leq & 1 \\
& & x_1, & & x_2, & & x_3, & & x_4, & & x_5, & & x_6 & \geq & 0
\end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

For the LP^1_{KP} problem, the split item is $s = 3$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{2}{3}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$

the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 8 \cdot \frac{2}{3} = \frac{49}{3}$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 11 ($LB(\bar{\boldsymbol{x}}) = 11$).

Due to the current Lower Bound LB and Upper Bound UB, we have:

$$11 \quad \leq \quad z(\, ILP\,) \quad \leq \quad \underbrace{\left\lfloor \frac{49}{3} \right\rfloor}_{=16}$$

- **Iteration 1**

The LP^1_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_3. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

(a) Solution of the LP^2_{KP} problem

We have

$$\mathscr{I}^2 = \{1, 2, 4, 5, 6\}, \quad \mathscr{I}^2_{\text{out}} = \{3\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

The capacity is $c(2) = 9$ and the objective-function constant is $p(2) = 0$. The split item is $s = 4$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_4 = \frac{4}{7}, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0 \qquad \text{and}$$

$$z(\, LP^2_{\text{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 9 \cdot \frac{4}{7} = \frac{113}{7}$$

The LP^2_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^2_{\text{KP}}) \right\rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{113}{7} \right\rfloor}_{=16} > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^2_{KP} problem is **inserted in the problem list** L.

(b) Solution of the LP^3_{KP} problem

We have

$$\mathscr{I}^3 = \{1,2,4,5,6\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{3\}$$

The capacity is $c(3) = 3$ and the objective-function constant is $p(3) = 8$. The split item is $s = 2$ and the residual capacity is $\bar{c} = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

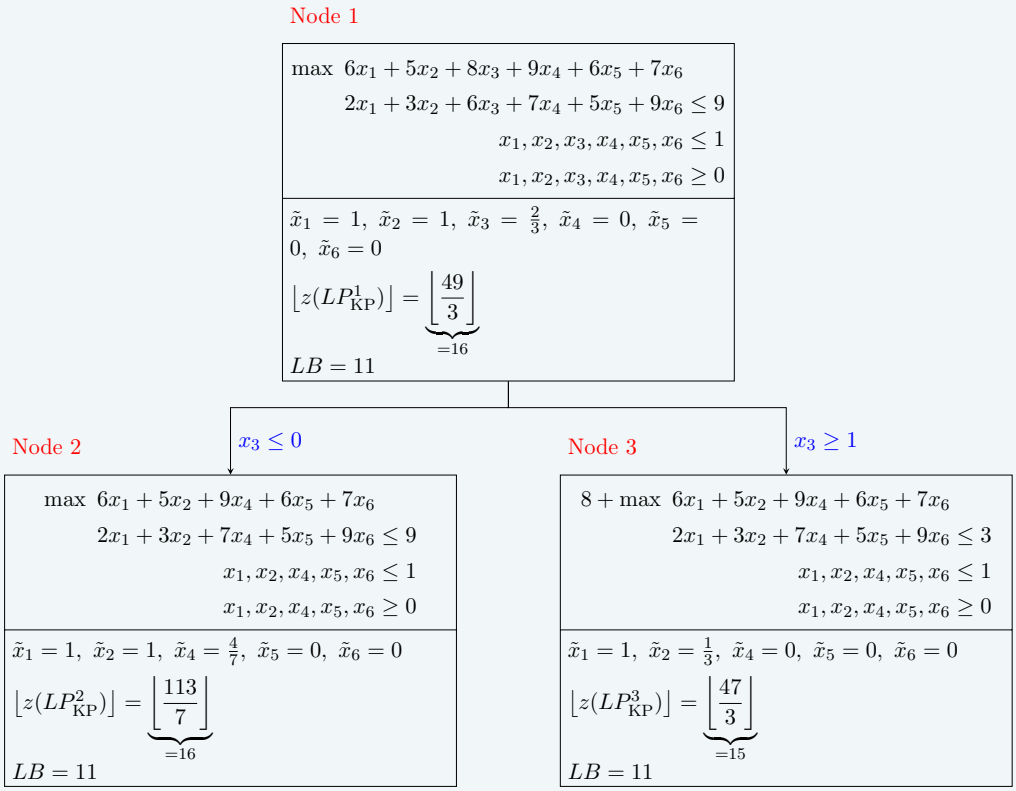
$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{1}{3}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0 \qquad \text{and}$$
$$z(\, LP^3_{\text{KP}}\,) = 8 + 6 \cdot 1 + 5 \cdot \frac{1}{3} = \frac{47}{3}$$

The LP^3_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^3_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{47}{3} \right\rfloor}_{=15} > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^3_{KP} problem is **inserted in the problem list** L.

At the end of the first iteration, Node 2 and Node 3 are both inserted in the Problem List L. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$11 \;\leq\; z(\,ILP\,) \;\leq\; \underbrace{\max\left\{\underbrace{\left\lfloor\frac{113}{7}\right\rfloor}_{=16},\; \underbrace{\left\lfloor\frac{47}{3}\right\rfloor}_{=15}\right\}}_{=16}$$

- **Iteration 2**

The LP^2_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_4. The LP^4_{KP} and LP^5_{KP} problems are created by unselecting ($x_4 \leq 0$) and selecting ($x_4 \geq 1$) item 4.

(a) Solution of the LP^4_{KP} problem

We have

$$\mathscr{I}^4 = \{1,2,5,6\}, \quad \mathscr{I}^4_{\text{out}} = \{3,4\}, \quad \mathscr{I}^4_{\text{in}} = \emptyset$$

The capacity is $c(4) = 9$ and the objective-function constant is $p(4) = 0$. The split item is $s = 5$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,LP^4_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \;\; \tilde{x}_2 = 1, \;\; \tilde{x}_5 = \frac{4}{5}, \;\; \tilde{x}_6 = 0 \quad \text{and} \quad z(\,LP^4_{\text{KP}}\,) = 6\cdot 1 + 5\cdot 1 + 6\cdot\frac{4}{5} = \frac{79}{5}$$

The LP^4_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^4_{\text{KP}})\right\rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor\frac{79}{5}\right\rfloor}_{=15} > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\,LP^4_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^4_{KP} problem is **inserted in the problem list** L.

(b) Solution of the LP^5_{KP} problem

We have

$$\mathscr{I}^5 = \{1,2,5,6\}, \quad \mathscr{I}^5_{\text{out}} = \{3\}, \quad \mathscr{I}^5_{\text{in}} = \{4\}$$

The capacity is $c(5) = 2$ and the objective-function constant is $p(5) = 9$. The split item is $s = 2$ and the residual capacity is $\bar{c} = 0$. An optimal solution $\tilde{\boldsymbol{x}}(\,LP^5_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \;\; \tilde{x}_2 = 0, \;\; \tilde{x}_5 = 0, \;\; \tilde{x}_6 = 0 \quad \text{and} \quad z(\,LP^5_{\text{KP}}\,) = 9 + 6\cdot 1 = 15$$

The LP^5_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^5_{\text{KP}})\right\rfloor > LB \qquad \text{since} \qquad 15 > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\,LP^5_{\mathrm{KP}}\,)$ is **integer**, accordingly, the lower bound LB becomes 15 and the incumbent solution $\bar{\boldsymbol{x}}(ILP)$ becomes $\bar{x}_1 = 1, \bar{x}_2 = 0, \bar{x}_3 = 0, \bar{x}_4 = 1, \bar{x}_5 = 0, \bar{x}_6 = 0$.

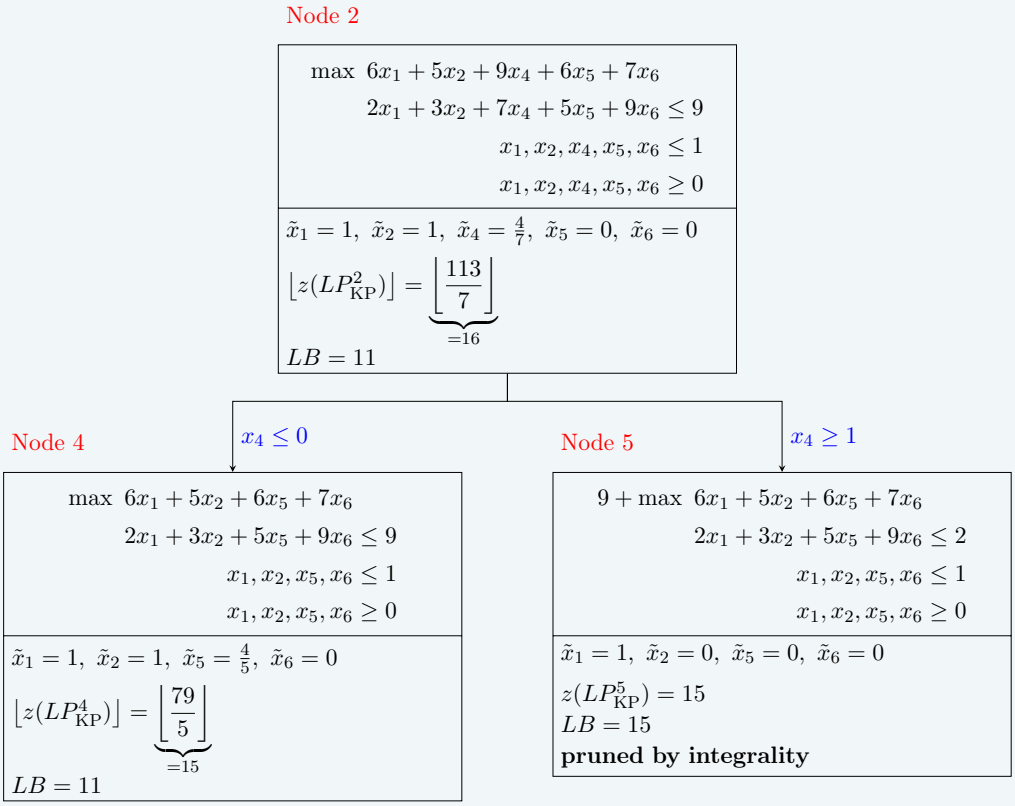
Accordingly, the branching node is pruned by integrality.

Since the Lower Bound is improved, the LP^3_{KP} problem and the LP^4_{KP} problem are pruned by upper bound and removed from the problem list L. We have:

$$\lfloor z(LP^3_{\mathrm{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{47}{3} \right\rfloor}_{=15} \leq 15$$

$$\lfloor z(LP^4_{\mathrm{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{79}{5} \right\rfloor}_{=15} \leq 15$$

At the end of the second iteration, Node 5 improves the Lower Bound to $LB = 15$ and is pruned by integrality; Node 3 and Node 4 are then pruned by upper bound. The corresponding Branch-and-Bound Tree is:



- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 15$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1, \tilde{x}_5 = 0, \tilde{x}_6 = 0$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.

